

A Fast Algorithm for Protein Structural Comparison

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Abstract

The goal of protein structural comparison attempts to establish an equivalence relation between polymer structures based on their shapes and three-dimensional conformations. Root mean square deviation (RMSD), a frequently-used approach, measures the average distance between the selected atoms of superimposed proteins. Although the RMSD is most popularly implemented, it suffers from a few drawbacks. For example, once the shapes of two proteins turn into divergent, RMSD loses its effectiveness and may result in high RMSD values. In this paper, we propose a simple method to compare protein structures by their spatial properties. First, protein chains are separated by Chameleon clustering. Then, each cluster represents a vertex and edges are determined by their geometric distances. Finally, a (undirected and unlabeled) protein graph is determined. Thus, the protein comparison problem becomes the problem of finding maximum common subgraph (MCS) for two protein graphs. However, the MCS problem is NP-hard. For efficiently finding MCS, we propose a simple heuristic algorithm according to degree sequences of subgraphs for estimating the size of MCS. Comparing with general RMSD approach, our method provides an alternative conception and promotive advantage on its efficiency. This graph-based approach offers a practical direction for protein structural comparison.

1 Introduction

With the increase of protein structures until 19 Apr 2011, 72550 structures are determined in the Protein Data Bank (PDB). Proteins are realized that they are indispensable materials to life because responsible for all vital reactions in a organism, including storage of energies, transmission of signals, and so on. In other words, if all of con-

trol mechanism of proteins function can be aware, it is helpful to prevent of disease and is useful for drug design. It is also known that a protein sequence determines its three-dimensional structure and the this three-dimensional structure determines its specific function. By computing structural similarity between two proteins, it is available to reveal some further information for protein function prediction and evolutionary relationships of proteins. The three-dimensional structure of a protein can be represented by secondary structure elements, the coordinates of all atoms in the protein, or even only the coordinates of C_α atom in each residue of protein. Therefore many structure alignment algorithms have been proposed for comparing proteins according to their specific structure representations.

Some proteins may have identical or similar structures but their sequence identities are very low [1]. This exception has a great influence to the identification of protein functions. Thus, protein structure comparison becomes an important research issue. There is no denying that if we want to understand their relationships of all protein structures, we probably need to perform comparisons in C_2^{65378} times. Therefore, our motivation is to propose a fast and efficient comparison algorithm to estimate similarity of two proteins. In other words, we are possible to consider two similar structural proteins but they are dissimilar by using sequence comparison methods like [2], [3], and [4].

Our purpose aims at the local structures of proteins to be an identification basis. Recurring substructures in proteins reveals important information about protein classification, functional prediction, and folding [5]. In this paper, we propose a graph-based approach to compare two protein structures. Not only obtains an allowable classification result for two different protein structural families in the SCOP database (Structural Classification of Proteins) [6], the proposed method also provides an efficient algorithm for estimating the similarity of two proteins.

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Table 1: The structural alignment algorithms based on C_α protein.

Name	Description
DALI [7]	A network service for comparing protein structures in 3D.
VAST [8]	A service that allows searching for structural neighbors.
CE [9]	Tools for 3-D protein structure comparison and alignment.
TopMatch [10]	A service for the alignment and superposition of structures.

2 Related Works

At Present, most of structural similarity comparison methods based on structural alignment are partitioned into two main types. One is to compare the two structures after superposition of aligned substructures, attempting to match the positions of corresponding residues [11]. Another type is to compare the distance matrices which records the intra-molecular distances between all residue pairs of two given protein structures respectively, and attempt to find an optimal match corresponding intra-molecular distances for selected aligned substructures [12].

There are two widely used similarity measures presented, that is, cRMSD (c Root Mean Square Deviation) and dRMSD (distance Root Mean Square Deviation). The main advantage of dRMSD is able to avoid wrong superimposition between two proteins so that we may obtain a larger value of RMSD although the two proteins are similar. Some of these methods compare respective distance matrices to each structure, trying to match corresponding intra-molecular distances for aligned substructures. Other methods compare the structures directly after superposition of aligned substructures, trying to match the positions of corresponding atoms.

In addition, TM-score [13] is an algorithm to calculate the similarity of topologies of two protein structures. It can be exploited to quantitatively access the quality of protein structure predictions relative to native. Because TM-score weights the close matches stronger than the distant matches, TM-score is more sensitive than RMSD. A single score between (0,1] is assigned to each comparison. Based on statistics, if a template/model has a TM-score around or below 0.17, it means the prediction is nothing more than a random selection from PDB library. Note that two completely unrelated proteins may have a large RMSD, but so may two related chains which consist of identical subunits oriented differently with respect to

each other. RMSD cannot distinguish the first case from the second one.

Therefore, many various searching algorithms used to obtain the minimum differences optimally have been proposed, having based on Dynamic programming [14], simulated annealing, genetic algorithms, Monte Carlo, geometric hashing [15], and graph theory in [16], and so on. [17] presented a structural alignment method based on bipartite graph matching to obtain a good match.

3 Methods

In this section, to explain our proposed algorithm, we start with a broad overview of our problem of structural comparison.

3.1 Problem Formations

In order to perform a structural comparison between molecules, it is required to obtain correct information from two superimposed protein structures. However, it is very difficult to optimize these two quantities simultaneously, since one can be optimized at the expense of the other [18]. Unlike the sequence alignment problem, the structural alignment problem has not been even classified as solvable. Proteins are made up of elements such as carbon, hydrogen, nitrogen, and oxygen. To be able to perform their biological function, proteins fold into specific spatial conformations, driven by a number of noncovalent interactions. For each atom in a protein, we simply adopt **PAM** (Partitioning Around Medoids) [19] and Chameleon clustering method, transforming a protein structure to an undirected simple graph. Figure 1 shows the idea of transformation. By doing so, the problem of protein structural comparison can be simplified as a graph problem shown in Figure 2. In such a manner, we expect the problem of protein structural comparison can be transferred to a basic issue, building by a graph-based

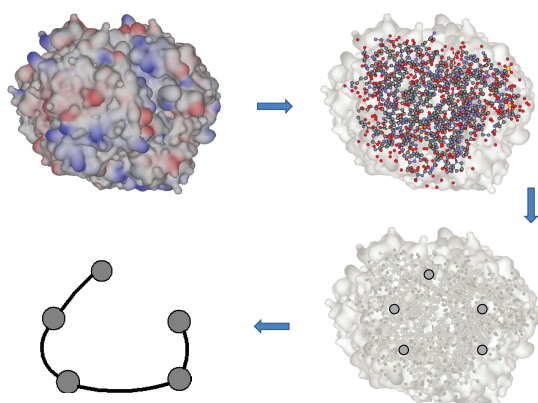


Figure 1: An illustration of protein structure remodeling.

method.

Given two proteins abbreviated graphs $G_A = (V_A, E_A)$ and $G_B = (V_B, E_B)$, let $V_A = \{a_1, \dots, a_m\}$ and $V_B = \{b_1, \dots, b_n\}$ such that $m \geq n$. The goal is to find a largest induced subgraph of G_B isomorphic to a subgraph of G_A , which is an MCS (maximum common subgraph) in their superimposing graph. It is known that finding MCS is an optimization problem and is NP-hard [20]. Hence, in the following section, we will develop a simple heuristic algorithm to estimate the size of MCS for the two protein graphs.

3.2 Graph-Based Protein Transformation

As mentioned in the remodeling of protein structure to graph, a consistent model is required to label each protein atom, converting into graph vertex. The data pre-processing work is performed by PAM (Partition Around Medoids) clustering method, partitioning data points into a set of K clusters. Comparing to K -mean clustering algorithm, PAM has the following features. First, it operates on the dissimilarity matrix of the given data set. Second, it is more robust because minimizing a sum of dissimilarities, instead of a sum of squared Euclidean distances. Third, it provides a novel graphical display. Whereas K -mean clustering may yield wrong clustering result, suffering from the influence of noises and outliers. Thus, the improved algorithm of PAM is adopted to refine atoms clustering.

Since partition clustering is hard to handle with non-spherical shape and arbitrary size, we use hi-

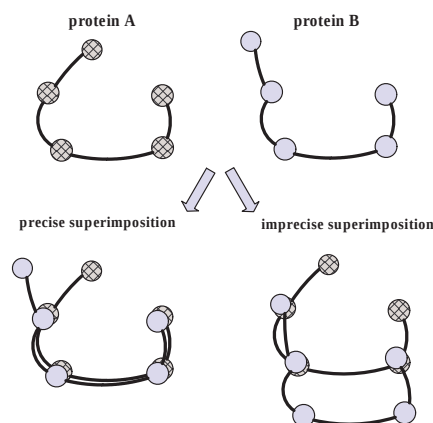


Figure 2: An illustration of protein superimposition.

erarchical clustering algorithm - **Chameleon**, to solve this restriction [21]. Here, a two-phase clustering algorithm is proposed to define the relation in protein reduced graph. Chameleon is a clustering algorithm using dynamic modeling presented to improve the weakness of CURE [22] and ROCK [23]. The algorithm can be divided into three major steps as shown in Figure 3. Initially, a k -nearest neighbor (K -NN) [24] graph is constructed to realize the relative relationship between each datum and its k nearest neighbors. Each vertex of a K -NN graph indicates a datum, and an edge between two vertices indicates that one is among the k nearest neighbors of the other. Then it uses a graph partitioning algorithm on the K -NN graph to yield a large number of small compact clusters and merges two small clusters satisfying that the inter-connectivity and closeness between two clusters are highly related to the internal inter-connectivity and closeness of data within the clusters repeatedly by an hierarchical agglomerative clustering algorithm.

Once the vertex set V of a graph is determined, the determination of the edge set E of a graph is significant to compute structural similarity. For this problem, we sum up all of distances of each pair of vertices of graph and calculate the average distance as a threshold of edges. The reason is due to that a edge can be compared to a bond connecting two atoms in a chemical structure. Because we discover that two atoms connected by a bond usually are adjacent or near to each other. Therefore, it is a basis to establish edges when the distance between two vertices is lower than the threshold.

Table 2: The partitioning clustering algorithms compared, MN is the number of maximal neighbors and NL is the number of local minima.

Algorithm	Time Complexity	Data Type	Input Required
K-means	$O(nkt)$	Numeric	k
PAM	$O(k(n-k)^2)$	Numeric	k
CLARA	$O(k(10-k)^2 + k(n-k))$	Numeric	k
CLARANS	$O(kn^2)$	Spatial	MN, NL

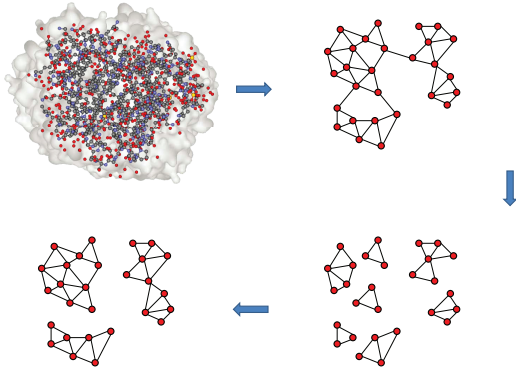


Figure 3: An illustration of Chameleon clustering algorithm.

3.3 Subgraph Isomorphism Problem

Given two undirected graphs transformed from protein three-dimensional structures, we propose a quantitative measure to compute the similarity between the two graphs. It is known that if two graphs are very similar, the size of their maximum common subgraph will be large. In other words, while the size of the maximum common subgraph of two graphs is large, their structural similarity is evaluated to be high. Therefore, our purpose is to find a maximum common subgraph of two graphs. The formal description of the graph isomorphism problem is defined as follows. Given two graph $G_A = (V_A, E_A)$ and $G_B = (V_B, E_B)$, if there is a bijective function f such that for any two vertices x and y of G_A with $(x, y) \in E$ if and only if $(f(x), f(y)) \in E_B$, we call G_A and G_B are isomorphic. However, the maximum common subgraph isomorphism problem is NP-complete. Therefore, we propose another alternative method to estimate the maximum common subgraph of two graphs.

The degree sequence of an undirected graph is a non-decreasing sequence of the degrees of vertices. Though it is known that two isomorphic graphs have the same degree sequence, two non-isomorphic graphs may also have the same degree sequence. However, it is ensured that two graphs with the same degree sequence, the sizes of the two graphs are equivalent. In our approach, we adopt the size of a graph to be a criterion for finding maximum common subgraph of two graphs, and the size of a graph G is the sum of the numbers of vertices and edges, *i.e.*, $|G| = |V| + |E|$.

The maximum subgraph of a graph is itself and therefore if the numbers of vertices of two compared graphs are distinct, we list all possible subgraphs of smaller graph, but only require to list the possible subgraphs of bigger graph whose number of vertices is no more than the number of vertices of smaller graph. Then, we calculate the degree sequences of these subgraphs, and the degree sequences containing in both graphs to be the candidates of maximum common subgraphs. In general, the number of candidates is more than one, so the maximum common subgraph is determined by calculating the sizes of these candidates.

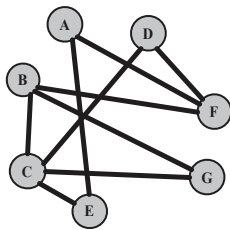
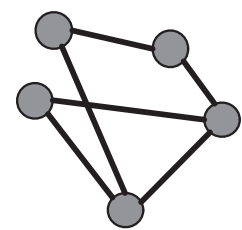
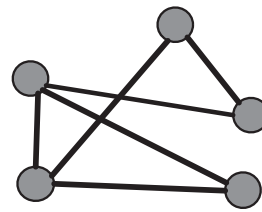
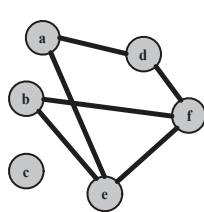
In the following, we show an example of graph comparison by our approach. First, two transformed graphs G_A and G_B are determined in Figure 4. We lists all possible subgraphs of G_A and G_B on different number of vertices, respectively. Tables 3 and 4 show the result. Then, we calculate the degree sequences of these subgraphs listed in Table 5, and find out the degree sequences containing in G_A and G_B . In this example, the equivalent degree sequences of G_A and G_B include 0, 00, 11, 000, 011, 112, 222, 0011, 0112, 1122, 1223, 2222, 0222, 01122, 01223, and 22233. Due to the number of candidates is more than one, so the maximum common subgraph G_C with the degree sequence 22233 is determined by comparing the sizes of all candidates, and the maximum common graph G_C of G_A and G_B is shown in Figure 5.

Table 3: Subgraphs of graph G_A .

V.N.	Graph G_A
1	A, B, C, D, E, F, G
2	AB, AC, AD, AE, AF, AG, BC, BD, BE, BF, BG, CD, CE, CF, CG, DE, DF, DG, EF, EG, FG
3	ABC, ABD, ABE, ABF, ABG, ACD, ACE, ACF, ACG, ADE, ADF, ADG, AEF, AEG, AFG, BCD, BCE, BCF, BCG, BDE, BDF, BDG, BEF, BEG, BFG, CDE, CDF, CDG, CEF, CEG, CFG, DEF, DEG, DFG, EFG
4	ABCD, ABCE, ABCF, ABCG, ABDE, ABDF, ABDG, ABEF, ABEG, ABFG, ACDE, ACDF, ACDG, ACEF, ACEG, ACFG, ADEF, ADEG, ADFG, AEFG, BCDE, BCDF, BCDG, BCEF, BCEG, BCFG, BDEF, BDEG, BDFG, BEFG, CDEF, CDEG, CDFG, CEFG, DEFG
5	ABCDE, ABCDF, ABCDG, ABCEF, ABCEG, ABCFG, ABDEF, ABDEG, ABDFG, ABEFG, ACDEF, ACDEG, ACDFG, ACEFG, ADEFG, BCDEF, BCDEG, BCDFG, BCEFG, BDEFG, CDEFG
6	ABCDEF, ABCDEG, ABCDFG, ABCEFG, ABDEFG, ACDEFG, BCDEFG

Table 4: Subgraphs of graph G_B .

V.N.	Graph G_B
1	a, b, c, d, e, f
2	ab, ac, ad, ae, af, bc, bd, be, cd, ce, cf, de, df, ef
3	abc, abd, abe, abf, acd, ace, acf, ade, adf, aef, bcd, bce, bcf, bde, bdf, bef, cde, cdf, cef, def
4	abcd, abce, abcf, abde, abdf, abef, acde, acdf, acef, adef, bcde, bcdf, bcef, bdef, cdef
5	abcde, abcdf, abcef, abdef, acdef, bcdef
6	abcdef

Figure 4: Two given protein graphs G_A and G_B .Figure 5: Maximum common subgraph of G_A and G_B .

Finally, a quantitative measure for computing graph similarity is defined as follows.

$$\delta = \frac{2(|V_C| + |E_C|)}{(|V_A| + |E_A|) + (|V_B| + |E_B|)} \quad (1)$$

where $G_A = (V_A, E_A)$ and $G_B = (V_B, E_B)$ are two graphs compared, and $G_C = (V_C, E_C)$ is the maximum common subgraph of G_A and G_B . In this formula, when the similarity between two graphs is

Table 5: Degree sequences of graphs G_A and G_B

V.N.	Graph G_A	Graph G_B
1	0	0
2	00, 11	00, 11
3	000, 011, 112, 222	000, 011, 112, 222
4	0112, 1122, 0222, 0011, 1113, 1111, 2222, 1223	0011, 0112, 1122, 1223, 2222, 0222
5	11123, 12223, 01223, 22222, 01111, 11222, 01122, 11224, 22233 , 11233	01122, 01223, 22233 , 02222
6	222233, 112224, 122333, 111223, 122223, 122234	022233

very high, then δ will near to 1. On the other hand, if two graphs are very dissimilar, the δ will near to 0. Therefore, in this example, the δ between the two graphs G_A and G_B is $\frac{2(5+6)}{(7+9)+(6+6)} = 0.785$.

Algorithm 1 Similarity Measure

- 1: **Input:** Two protein graphs, G_A and G_B .
 - 2: **Output:** The similarity (δ)
 - 3: Generate all possible subgraphs into SS_A and their corresponding degree sequences into DS_A ;
 - 4: Generate all possible subgraphs into SS_B and their corresponding degree sequences into DS_B ;
 - 5: Find candidate degree sequence set (CDS) from DS_A and DS_B ;
 - 6: The maximum common subgraph G_C of G_A and G_B is determined by calculating the size of the graphs of the CDS ;
 - 7: Compute the similarity (δ) of G_A and G_B ;
 - 8: **return** δ
-

4 Results

In order to demonstrate our approach that is useful to assess the similarity of protein structures, we have to examine some of the three-dimensional protein structure data from PDB. It is discovered that some proteins have similar three-dimensional structures, but their amino acid sequences are dissimilar. In [25], it describes that some similar protein structures, *e.g.*, myoglobins, cannot be detected by sequence alignment.

In this experiment, we take six similar structures of G proteins as input. Table 6 shows the

annotations of these six proteins. Table 7 shows the experimental result of our method. Table 8 shows the results obtained by RMSD. Let us examine the two similar proteins 1QRA and 1GNP first. By our approach, the structural similarity of the two proteins is 1.0000. However, it gets a score 0.4 by RMSD. As a result, our approach obtains a better result than RMSD approach.

5 Conclusion

In this paper, we give a simple approach to moderate protein structure by its spatial properties. Comparing with general RMSD approach and its ability, our method provides an alternative conception and promotive advantage on its efficiency. This graph-based approach offers a practical direction for protein structural comparison.

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Table 6: The annotation of the G proteins family.

Protein ID	1AA9	1GNP
Length	171	166
Class	Alpha and beta proteins (a/b)	Alpha and beta proteins (a/b)
Fold	P-loop containing nucleoside triphosphate hydrolases	
Superfamily	P-loop containing nucleoside triphosphate hydrolases	P-loop containing nucleoside triphosphate hydrolases
Family	G proteins	G proteins
Domain	cH-p21 Ras protein	cH-p21 Ras protein
Protein ID	1QRA	5P21
Length	166	171
Class	Alpha and beta proteins (a/b)	Alpha and beta proteins (a/b)
Fold	P-loop containing nucleoside triphosphate hydrolases	Immunoglobulin-like beta-sandwich
Superfamily	P-loop containing nucleoside triphosphate hydrolases	P-loop containing nucleoside triphosphate hydrolases Immunoglobulin
Family	G proteins	G proteins V set domains (antibody variable domain-like)
Domain	cH-p21 Ras protein	cH-p21 Ras protein

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Table 7: The annotation of the G proteins family.

Protein ID	1CD8	1NEU
Length	166	166
Class	All beta proteins	All beta proteins
Fold	P-loop containing nucleoside triphosphate hydrolases	Immunoglobulin-like beta-sandwich
Superfamily	Immunoglobulin	Immunoglobulin
Family	V set domains (antibody variable domain-like)	V set domains (antibody variable domain-like)
Domain	CD8	Myelin membrane adhesion molecule P0

Table 8: The similarity obtained by our method and RMSD.

PID	1AA9	1GNP	1QRA	5P21	1CD8	1NEU
1AA9	1.000 0.0	0.848 1.7	0.848 1.9	1.000 1.7	0.750 4.0	0.807 3.7
1GNP	-	1.000 0.0	1.000 0.4	0.848 0.4	0.612 2.4	0.845 3.5
1QRA	-	-	1.000 0.0	0.848 0.4	0.612 3.6	0.845 3.2
5P21	-	-	-	1.000 0.0	0.750 3.2	0.807 2.6
1CD8	-	-	-	-	1.000 0.0	0.577 2.3
1NEU	-	-	-	-	-	1.000 0.0

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